

Threshold Gates for Activation Sparsity: Early Fixed-sReLU Evidence and Learnable Ablation Plan

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- **Working name.** Use *Adaptive Threshold Gates* (ATGs) as the umbrella name. ATG^+ is a one-sided thresholded ReLU; $\text{ATG}^\pm$ is the signed magnitude gate currently called sReLU. “Adaptive” is reserved for learned κ ; fixed- κ gates are fixed ATGs.
- **Early AdamW result.** Replacing ordinary POST Q/K/V ReLUs with fixed $\text{ATG}_{0.1}^\pm$ lowers validation loss from 5.0991 to 4.9829, but lowers R_{model} from 12.91% to 12.12%. Signed-value preservation helps this single-run endpoint; a fixed absolute threshold does not prevent scale evasion.
- **Early OR result.** With the inherited $c = \sigma = 0.05$, OR reduces R_{model} from 12.12% to 6.09% on fixed $\text{ATG}_{0.1}^\pm$. This is a geometric incompatibility: $\sqrt{3}c = 0.0866 < \kappa$, so every nonzero survivor begins outside the Ricker attraction basin.
- **Learnable does not mean directly differentiable.** A hard comparison has zero κ -gradient almost everywhere. A learned ATG therefore needs an explicit estimator, checkpointed threshold parameters, and a sparsity/compute incentive; task loss alone is expected to favor smaller thresholds.
- **Recommended program.** Use a matched 2,048-step, 134.2M-token screen, promote Pareto candidates to 8,192 steps, then run full-pass multi-seed confirmations. Do not run a full Cartesian product or compare partial endpoints with full-pass endpoints.
- **Evidence boundary.** All reported training endpoints are seed-0, random-initialized Pythia-14M pre-training over one MiniPile token-cache pass. Logical zero products are not measured speedups. Config 119 is provisional because its terminal artifact publication failed after training and validation had finished.

1 Gate Family and Placement

For a scalar activation x and nonnegative threshold κ , define

$$G_\kappa^+(x) = x \mathbf{1}[x \geq \kappa], \quad G_\kappa^\pm(x) = x \mathbf{1}[|x| \geq \kappa]. \quad (1)$$

G_0^+ is ordinary ReLU. $G_0^\pm$ is the identity apart from signed zero, not ReLU. The harness name `symmetric_threshold` and the shorthand “sReLU” refer to $G^\pm$, but *symmetric magnitude gate* is mathematically clearer because negative survivors remain negative.

The proposed nomenclature is:

- **fixed ATG^+ / fixed $\text{ATG}^\pm$:** κ is a hyperparameter;
- **learned ATG^+ / learned $\text{ATG}^\pm$:** κ is optimized;
- **scope:** always name the gated sites, for example POST-QKV learned $\text{ATG}^\pm$ or Three-ReLU learned ATG^+ .

This is working project nomenclature, not a novelty claim.

The completed fixed-threshold experiment keeps ordinary ReLU at the three branch sites $a_l = \text{ReLU}(\text{LN}_l^{\text{att}}(H_l))$, $m_l = \text{ReLU}(\text{LN}_l^{\text{mlp}}(H_l))$, and $h_l = \text{ReLU}(m_l W_{1,l} + b_{1,l})$. It changes only the post-QKV gates:

$$\begin{aligned} [Q_l^0, K_l^0, V_l^0] &= a_l W_{\text{qkv},l} + b_{\text{qkv},l}, & K_l &= G_{0.1}^\pm(\text{RoPE}(Q_l^0)), & V_l &= G_{0.1}^\pm(V_l^0), \\ Q_l &= G_{0.1}^\pm(\text{RoPE}(Q_l^0)), & P_l &= \text{softmax}(Q_l K_l^\top / \sqrt{d_h} + \mathcal{M}), & C_l &= P_l V_l. \end{aligned}$$

Thus Q/K are gated after RoPE and V immediately before PV. Future learned ATG⁺ tests may also replace the ordinary ReLUs at a_l, m_l, h_l ; those are distinct architectures and require matched controls.

2 Hypotheses

1. **Signed preservation.** ATG[±] can retain informative negative Q/K/V values that ordinary ReLU removes, improving quality at comparable logical compute.
2. **Site heterogeneity.** Q, K, V, attention/MLP inputs, and MLP hiddenes have different scales and loss sensitivities. Per-layer, per-site thresholds should dominate one global absolute κ .
3. **Scale evasion.** A fixed absolute cutoff can be bypassed by moving surviving magnitudes outside $[-\kappa, \kappa]$. RMS-relative thresholds or an explicit active-mask cost may be necessary.
4. **Pressure compatibility.** L1 can pull survivors across a hard boundary. Ricker pressure can do so only when its inward-gradient basin overlaps the survivor region; RN and OR share this scalar-field requirement even though their optimizer coupling differs.
5. **Compute-aware learning.** Optimizing average activation sparsity is not equivalent to optimizing R_{model} . Gate learning should eventually weight QKV, valid-causal QK/PV, W_o , W_1 , and W_2 by their logical product counts.
6. **Partial-budget predictiveness.** A 2,048-step screen should expose collapse and gross incompatibility, but candidate ranking must be backtested and rechecked at longer budgets.

3 Completed Full-Pass Evidence

Every row uses the same random-initialized 14,067,712-parameter Pythia-14M architecture, seed 0, 22,762 optimizer steps, 1,491,730,432 training tokens, and full validation on 692,224 tokens. OR uses $w = 1$, $c = \sigma = 0.05$, and step budget $b = 0.5$; OL1 uses $w = 5$ and $b = 0.5$. Fixed gates add no parameters.

Table 1: Training endpoints. The closest causal contrasts are 118 versus 110, 119 versus 111, and 119 versus 118. Configs 50, 77, and 98 are progressively more distant architecture anchors.

Cfg.	Architecture	Method	Q/K/V gate	κ	Val. loss	Time (h)	Evidence
50	Stock GELU	AdamW	none	—	4.8317	3.010	complete
77	One-ReLU	AdamW	none	—	4.8404	2.897	complete
98	Three-ReLU	AdamW	none	—	4.9564	3.449	complete
110	Six-ReLU POST	AdamW	G^+	0	5.0991	3.455	complete
111	Six-ReLU POST	OR	G^+	0	5.1358	4.212	complete
112	Six-ReLU POST	OL1	G^+	0	5.1731	4.087	complete
118	POST fixed ATG [±]	AdamW	$G^\pm$	0.1	4.9829	3.249	complete
119 [†]	POST fixed ATG [±]	OR	$G^\pm$	0.1	5.0122	4.275	provisional

Throughput (tokens/s), in row order: 137,648; 143,021; 120,139; 119,932; 98,389; 101,399; 127,520; 96,925. Dense PyTorch executes the same matmuls, so these differences are architecture/method overhead and run variability, not sparse acceleration.

An activation is counted as exactly zero only when its computed floating-point value satisfies $x = 0$; there is no tolerance. Table 2 pools integer counts across all six layers, all 338 complete validation sequences, all 692,224 evaluated tokens, and every relevant feature/head coordinate. For QK and PV, only valid causal pairs enter the product denominators.

Table 2: Full-validation exact-zero and logical-product outcomes. The two zero-rate columns report $z_a/z_m/z_h$ and $z_Q/z_K/z_V$, respectively; Q/K/V are the actual POST QK/PV operands. R_{block} counts logical zero products inside transformer blocks. R_{model} uses the same numerator and adds the dense LM head to the denominator. Neither is latency.

Cfg.	Architecture	Method	Val. loss	$z_a/z_m/z_h$	$z_Q/z_K/z_V$	R_{block}	R_{model}
50	Stock	AdamW	4.8317	—/—/—	—/—/—	0.00%	0.00%
77	One-ReLU	AdamW	4.8404	—/—/52.06%	—/—/—	7.44%	2.23%
98	Three-ReLU	AdamW	4.9564	49.54/50.14/50.72%	—/—/—	19.71%	5.90%
110	POST ReLU	AdamW	5.0991	48.52/50.06/50.11%	15.68/23.18/49.24%	43.12%	12.91%
111	POST ReLU	OR	5.1358	48.56/49.82/50.19%	64.13/65.77/50.39%	55.03%	16.48%
112	POST ReLU	OL1	5.1731	48.54/49.72/50.37%	81.21/81.39/51.33%	59.50%	17.82%
118	POST fixed ATG [±]	AdamW	4.9829	48.93/49.95/50.99%	5.46/10.84/57.15%	40.48%	12.12%
119 [†]	POST fixed ATG [±]	OR	5.0122	49.37/50.28/48.06%	0.75/0.79/2.01%	20.34%	6.09%

Table 3: Matched deltas; positive ΔR_{model} means more potentially avoidable logical products. Config 119 comparisons remain provisional.

Contrast	$\Delta \mathcal{L}_{\text{val}}$	ΔR_{block}	ΔR_{model}	Readout
Fixed ATG $^{\pm}$ vs ReLU, AdamW (118–110)	−0.1162	−2.64 pp	−0.79 pp	Better loss, slightly less compute opportunity
OR vs AdamW, POST ReLU (111–110)	+0.0366	+11.91 pp	+3.57 pp	OR increases ordinary-ReLU sparsity
OR vs AdamW, fixed ATG $^{\pm}$ (119–118)	+0.0293	−20.14 pp	−6.03 pp	OR drives symmetric-gate survivors outward
Fixed ATG $^{\pm}$ vs ReLU, OR (119–111)	−0.1235	−34.69 pp	−10.39 pp	Better loss, much lower logical sparsity

The clean AdamW contrast supports signed preservation as a quality hypothesis, but not as a sparsity result: Q/K zeros fall, V zeros rise, and net R_{model} falls by 0.79 points. The fixed threshold is also scale-sensitive: $|x| \geq 0.1$ is a local numerical condition, not a guaranteed density target.

Artifact boundary. Config 118 is completed. Config 119 reached all 22,762 steps, saved metrics and its final checkpoint, and completed the 692,224-token validation. Its manifest is nevertheless terminal **failed**: the atomic predictions temporary path was 261 characters on Windows, so `predictions.jsonl` was not published. Diagnostic config 120 explicitly records `source_manifest_status: failed`; this report does not repair or promote that run. The pinned IDs are 001-20260717-211157-e6f8dda0, 001-20260718-002702-ebfeb045, and diagnostic 001-20260718-104753-25aa16e9.

4 Why the Inherited OR Setting Is Incompatible

For activation u , the Ricker score and minimized pressure are

$$f(u) = \left(1 - \frac{u^2}{c^2}\right) \exp\left(-\frac{u^2}{2\sigma^2}\right), \quad \mathcal{P}(u) = 1 - f(u).$$

When $c = \sigma$, gradient descent on $\mathcal{P}$ moves values inward only for $|u| < \sqrt{3}c$ and outward for $|u| > \sqrt{3}c$. In config 119,

$$\sqrt{3}c = \sqrt{3}(0.05) = 0.0866 < \kappa = 0.1.$$

Every nonzero ATG $_{0.1}^{\pm}$ output therefore begins in the outward region. Values inside the dead zone are exactly zero and the hard gate blocks the ordinary activation gradient there. The observed aggregate Q/K/V exact-zero rate falls from 47.75% at step 1 to 2.14% by step 1,000 and 1.22% in the final training minibatch; full validation gives 1.18%. AdamW instead ends at 23.63% in the final minibatch and 24.48% on full validation.

This is evidence against the inherited pair $(\kappa, c) = (0.1, 0.05)$, not against Ricker pressure in general and not an OR-versus-RN conclusion. A compatible Ricker test must predeclare c, σ relative to κ and the observed activation scale; for $c = \sigma$, at minimum the survivor boundary must overlap $|u| < \sqrt{3}c$. OL1 is the cleaner next pressure control because it has no finite attraction basin.

5 Making the Threshold Learnable

Simply wrapping κ in a trainable parameter is insufficient: the indicator in Equation 1 has zero derivative with respect to κ almost everywhere. The first implementation should use a hard-forward, soft-backward mask. Let $v = x$ for ATG $^+$ and $v = |x|$ for ATG $^{\pm}$:

$$\begin{aligned} m_h &= \mathbf{1}[v \geq \kappa], & m_s &= \text{sigmoid}\left(\frac{v - \kappa}{\tau}\right), \\ m &= m_s + \text{stopgrad}(m_h - m_s), & y &= xm. \end{aligned}$$

The forward pass remains exactly sparse; the backward pass supplies a local κ gradient. Parameterize $\kappa = \text{softplus}(\rho)$ in FP32, exclude ρ from weight decay, and start with one scalar per layer/site. POST Q/K/V then adds only 18 threshold scalars.

The implementation contract is:

- $\log \kappa$, its gradient/step norm, transition-band mass, and the pre-gate margin $v - \kappa$ by layer/site;
- preserve threshold parameters and metadata through final checkpoint save/reload, then require output and exact-zero equality after a round trip;
- keep a distinct threshold-parameter optimizer group so its learning-rate multiplier survives scheduling;
- compare absolute κ with an RMS-relative variant after the basic path is verified;
- do not call task-only threshold learning a sparsity objective. Task loss will often shrink κ to recover signal; add an explicit soft active-mask/compute cost or a target-density dual as a separate method when needed.

Report 05’s site-specific and model-compute post-hoc clipping frontiers (Figures 101–102) already motivate nonuniform thresholds: Q, K, V, and the branch/MLP gates tolerate different cutoffs. They are ordinary-ReLU checkpoint diagnostics, not learned-ATG results.

6 Partial-Run Screening Design

The proposal is a *fixed token budget on the existing full MiniPile cache*, not a permanently reduced document subset. Keep the same 65,536 tokens/step, sampler seed, warmup, batch/accumulation settings, and full 692,224-token final validation for every matched row. This reduces cost without changing the data source. Partial and full-pass endpoints must never be mixed in one numerical ranking.

Table 4: Staged screen. Counts are upper bounds; later stages depend on promotion.

Stage	Steps/tokens	Runs	Variables	Purpose / expected cost
Engineering pilot	128 / 8.39M	few	τ ; κ LR; checkpoint round trip	Plumbing only; never scientific evidence
AdamW screen	2,048 / 134.22M	14	Three matched controls; fixed $\text{ATG}^+_{.03,.10}$; fixed $\text{ATG}^\pm_{.03,.05,.10,.20}$; learned ATG^+ at 1/3/6-gate scopes; learned POST-QKV $\text{ATG}^\pm$ initialized at .03/.10	About 4.2 GPU-h total
Pressure screen	2,048 / 134.22M	≤ 8	L1N, OL1, RN, OR on at most two Pareto gate candidates; geometry-aligned Ricker	About 3–4.5 GPU-h
Longer rung	8,192 / 536.87M	≤ 4	Rerun promoted candidates from scratch with matched controls	About 5–6.5 GPU-h
Confirmation	22,762 / 1.492B	≤ 2	Full-pass finalists; at least three seeds	Roughly 3–4.5 h per run

The 2,048-step budget is the established harness screen: 9.0% of one full pass. Existing throughput implies about 17.5–18.7 minutes for POST AdamW and 22–23 minutes for OR/OL1; naive pressure may be slower. It is long enough to expose the observed incompatibility—fixed $\text{ATG}^\pm$ /OR had already fallen to about 1% aggregate gate zeros by step 2,000—but not long enough to establish the final ordering.

Before launching the grid, backtest whether validation-loss and R_{model} rankings around step 2,000 agree with the available full-pass runs. Use successive halving rather than a Cartesian product. Hold model learning rate and effective batch size fixed in the primary screen; vary them one axis at a time only for a promoted gate/method pair.

Proposed promotion rule. Require a valid artifact, finite training, exact learned- κ reload, and membership on the matched-budget ($\mathcal{L}_{\text{val}}, R_{\text{model}}$) Pareto frontier. As a permissive screen, pre-register $\Delta\mathcal{L}_{\text{val}} \leq +0.05$ versus the same-topology AdamW control and require either at least +1 percentage point R_{model} or a clear loss gain without lower R_{model} . Also reject universal $\kappa \rightarrow 0$, runaway/all-zero gates, or sharply declining sparsity over the final quarter. These are screening rules, not paper claims.

7 Prioritized TODOs

1. **Implement learned ATGs safely:** STE/relaxation, FP32 threshold group, checkpoint reconstruction, metrics, and tests.
2. **Run the AdamW fixed- κ ladder:** especially $\text{ATG}^\pm$ at 0.03, 0.05, 0.10, and 0.20 with matched 2,048-step controls.
3. **Run learned thresholds:** $\text{ATG}^\pm$ on POST Q/K/V and ATG^+ on the One-, Three-, and Six-gate scopes; compare global, per-site, and per-layer/site κ only after the basic path works.

4. **Add pressure controls:** OL1 first, then L1N; run RN/OR only with a Ricker basin aligned to κ . If the orthogonal correction cap is active almost always, vary step budget before nominal pressure weight.
5. **Ablate attention scope:** Q-only, K-only, V-only, QK, and QKV; then revisit PRE versus POST RoPE for the best threshold family.
6. **Add a direct compute incentive:** soft active-mask cost or target- R_{model} dual if task-only learned κ collapses toward zero.
7. **Tune after promotion:** threshold learning rate/temperature, model learning rate, RN/OR/L1 weights and budgets, then effective batch size. Do not cross all axes at once.
8. **Confirm robustness:** longer budgets, at least three seeds, larger Pythia models, and measured latency only after a compatible sparse kernel exists.
9. **Defer probability gating:** softmax P is positive, so ordinary ReLU is inactive there; retain learnable positive probability thresholding as a separate TODO.

8 Current Recommendation

Proceed with the staged partial-budget design. The immediate scientific priority is not another full-pass run: it is to determine whether the early fixed-ATG behavior is stable across κ , whether learned thresholds can remain nontrivial under a well-defined gradient estimator, and whether OL1 improves the quality- R_{model} frontier. Keep the inherited OR result as a useful negative control, then rerun Ricker only after its basin is made compatible with the gate boundary.